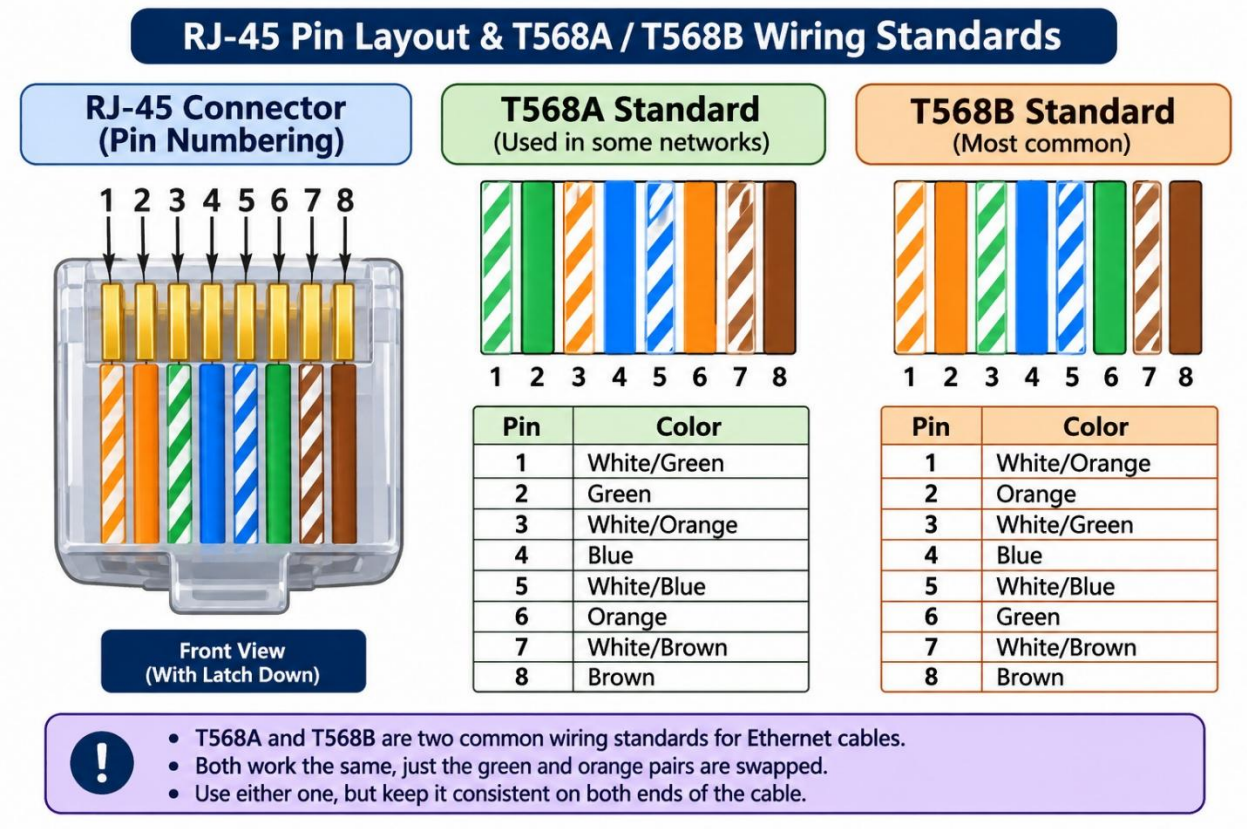
Ethernet Standard and Specification

Task 1 : Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

- An RJ-45-style 8P8C connector has eight pins used to terminate the eight conductors of an Ethernet twisted-pair cable. T568A and T568B are two common wiring standards.



Task 2 : Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Ethernet Standard	Speed	Cable Type	Maximum Cable Length	Connector Type
10BASE-T	10 Mbps	Cat3 or better twisted-pair cable	100 metres	RJ-45-style 8P8C
100BASE-TX	100 Mbps	Cat5 or better twisted-pair cable	100 metres	RJ-45-style 8P8C
1000BASE-T	1 Gbps (1000 Mbps)	Cat5e or better twisted-pair cable	100 metres	RJ-45-style 8P8C

Explanation

1. **10BASE-T:** An older Ethernet standard that supports speeds of up to 10 Mbps.
2. **100BASE-TX:** Fast Ethernet supports speeds of up to 100 Mbps.
3. **1000BASE-T:** Gigabit Ethernet supports speeds of up to 1 Gbps and is commonly used in modern home and office networks.

Task 3 : Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

- If I need to connect two laptops directly using Ethernet, a **crossover cable** is the traditional choice when their network interfaces do not support Auto-MDI/MDI-X.
- A crossover cable uses T568A at one end and T568B at the other end. This swaps the transmit and receive pairs for compatible 10/100 Mbps Ethernet interfaces.
- However, most modern laptops and Ethernet adapters support Auto-MDI/MDI-X, which automatically adjusts the transmit and receive pairs. Therefore, a straight-through cable will usually work with modern devices as well.

Task 4 : List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Device	Likely Ethernet Standard	Usage
Desktop Computer	1000BASE-T	Connects to the network for internet access, online gaming, and file transfers.
Wi-Fi Router (LAN Port)	1000BASE-T	Connects wired devices to the local network and internet.
Network Switch	1000BASE-T	Connects multiple computers and other network devices within a LAN.

Explanation

1. **Desktop Computer:** A Gigabit Ethernet port provides a fast and stable wired network connection.
2. **Wi-Fi Router:** Its LAN ports allow computers, smart TVs, and other devices to connect through Ethernet.
3. **Network Switch:** Gigabit Ethernet ports allow multiple devices to communicate efficiently on a local network.

Note: The actual Ethernet standard depends on the device model and its port specifications.